

Large Language Models (LLMs) — Quick Notes

- **Definition:** Deep learning models trained on massive text corpora to understand and generate human-like language.
- **Architecture:** Based on the Transformer, using self-attention to weigh relationships between all tokens in a sequence.
- **Training:** Pre-trained via next-token prediction (unsupervised), then fine-tuned using instruction tuning and RLHF for alignment.
- **Scale:** Performance improves with more parameters, data, and compute — often described by scaling laws.
- **Tokenization:** Text is broken into subword tokens (e.g., BPE) before being processed by the model.
- **Applications:** Chatbots, summarization, translation, code generation, question answering, and more.
- **Limitations:** Can hallucinate facts, reflect biases in training data, and require significant compute for training/inference.
- **Examples:** GPT, Claude, LLaMA, Gemini, Mistral.